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The Raven's Advanced Progressive Matrices: a comparison of relationships with verbal ability tests

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This study examined relationships between the Raven's Advanced Progressive Matrices, the Similarities subtest of the South African Wechsler Adult Intelligence Scales and an adaptation of the Stanford Diagnostic Reading Test Reading Comprehension subtest. Comparisons between these relationships were drawn on the basis of home language and gender in a sample of 100 university students. Results indicated significant relationships between the Advanced Matrices and both verbal tests, with no significant differences between the correlations on the basis of either gender or home language. This suggests that convergent validity is supported across these groups, and provides impetus for future research about the suitability of the Advanced Matrices for use within cross-cultural, multi-lingual contexts such as South Africa.

Keywords: construct validity; Ravens Advanced Progressive Matrices; verbal-analyticability

The Raven's Progressive Matrices epitomize one of the first and most successful attempts to present inductive reasoning and analogical tasks in non-verbal format. The linguistically minimized nature of the tests is particularly important, as it theoretically allows for an evaluation of fluid intelligence without substantial influence by language, educational and cultural factors. Consequently, the Ravens Matrices are considered to be a less biased and fairer measure of cognitive functioning across different populations, explaining their popularity throughout the world in both psychological practice and research, particularly within a multi-cultural context such as South Africa (Raven, 1989; Raven, Raven & Court, 1998).

It is particularly useful to investigate the Raven's Advanced Progressive Matrices (hereafter Advanced Matrices) in the South African context for several reasons, not least of which is the test's wide-spread use in both applied and research psychology as an easy-to-administer, group or individual assessment of intellectual functioning (Carpenter, Just, & Shell, 1990; DeShon, Chan, & Weissbein, 1995). Further, the test contains a relatively large number of items, making it particularly suitable for in-depth statistical analysis; and a large database of performance profiles and norms from numerous different populations and countries is available, enabling high levels of cross-cultural comparison (Carpenter *et al.*, 1990). Due to their simplicity and non-verbal nature, the Advanced Matrices are also often promoted as more culturally fair than other common measures of intellectual performance, implying the test could potentially be suitable for use across cultural, linguistic and racial groups within South Africa (Owen, 1992).

Since their development, much research has focused on the constructs underlying performance on the Advanced Matrices. Several factor analytic studies have indicated a single latent variable responsible for performance, which has been identified as that aspect of Spearman's *g* which taps educative ability and reasoning, and, in the case of the Advanced Matrices, is "a test of a person's capacity to form comparisons, reason by analogy, and develop a logical method of thinking, regardless of previously acquired information" (Raven, 1938, p.12). Raven, Court and Raven (1977) cautioned that the Advanced Matrices are not a pure measure of *g* as they exclude an important aspect, namely verbal ability.

Many researchers suggest that the Advanced Matrices include heterogeneous problem-solving processes, for example, both visual-spatial and verbal reasoning. In support of this, several factor analytic studies have found multiple factors on the Advanced Matrices. Most suggest that the reasoning ability assessed by the Advanced Matrices entails at least two aspects of analogical reasoning,

namely a verbal-analytical factor (Carroll, 1993; DeShon *et al.*, 1995; Lynn, Allik, & Irwing, 2004; McGrew & Flanagan, 1998) and a visualization and/or spatial ability factor (Colom & Garcia-Lopez, 2002; DeShon *et al.*, 1995; Dillon, Pohlman & Lohman, 1981; Lynn *et al.*, 2004). Lynn *et al.*'s (2004) factor analysis of the Standard Progressive Matrices (hereafter Standard Matrices) revealed three factors (gestalt continuation, verbal-analytic reasoning and visuo-spatial ability), plus a higher order factor, identified as *g*, supporting Carroll's (1997) statement that *g* typically constitutes more than half of the total common factor variance in tests of cognitive ability. The verbal element measured in the Advanced Matrices appears to be those aspects of *g* that involve using rules for analysis, coding and transforming relationships, which are also required for the evaluation of non-verbal content (Olson, 1986).

Most studies have focused on the visuo-spatial and non-verbal reasoning components of the Advanced Matrices, while far fewer have examined the verbal-analytic construct validity of its scores. Given the extensive use of the Advanced Matrices in the measurement of cognitive abilities, it is important that the processing components underlying the test are understood. Consequently, the current study investigated the relationship between scores on the Advanced Matrices and on two verbal measures, namely the Similarities subtest of the South African Wechsler Adult Intelligence Scale (SAWAIS, based on the Wechsler Adult Intelligence Scale), a measure of verbal-analytical reasoning (Lezak, Howieson, & Loring, 2004), and the Reading Comprehension subtest of the Stanford Diagnostic Reading Test (SDRT), a measure of English language comprehension (Spreen & Strauss, 1998).

The study also compared the above relationships across gender and home language groups. Limited research exists with regard to the effect that 'mother-tongue' may have on Advanced Matrices performance, despite evidence suggesting that language has statistically significant effects on measurements of cognitive functioning (Raven *et al.*, 1998; Strauss, 2003). In comparison, the role gender may play in determining Advanced Matrices performance is far better researched, but controversial. Much of the interest in gender differences in cognitive ability stems from research that points to a male advantage in spatial ability and a female advantage in receptive and expressive verbal ability (Hyde, 1981; Hyde, Fennema, & Lamon, 1990; Hyde & Linn, 1988; Neisser *et al.*, 1996). Despite this, numerous studies suggest that there are no statistically significant gender differences in performance on the Advanced Matrices (e.g. Court & Kennedy, 1976; Jensen, 1998; Mackintosh, 1996; Rushton, Skuy, & Fridjhon, 2003; Thissen, 1976). Court (1983), on the basis of an extensive review of studies on the various Ravens Matrices, concluded that "accumulated evidence at all ability levels indicates that a biological sex difference cannot be demonstrated" (p.68).

The evidence, however, is mixed as a range of studies have found male-female differences on the Advanced Matrices (e.g. Arthur & Woehr, 1993; Heron & Chown, 1967, Lynn, 2002; Lynn & Irwing, 2004; Paul, 1985). Lynn (1999; 2002) and Lynn *et al.* (2004) found a statistically significant (2- to 5-point mean IQ) gender difference in performance on the Advanced Matrices favouring males, while a factor analytic study by Lim (1994) suggested that the Advanced Matrices measure different abilities in males and females (reasoning ability in males compared to reasoning and spatial ability in females). Following this line of reasoning, Abad, Colom, Rebollo, and Escorial (2004) proposed that the visuo-spatial nature of many of the items on the Advanced Matrices is biased against females. Colom, Escorial, and Rebollo's (2004) findings support this as the male advantage that they found on the Advanced Matrices was non-significant when differences in spatial ability were controlled for. To complicate matters, Lynn *et al.* (2004) found that on all four of their identified factors on the Standard Matrices, girls outperformed boys at age 12, while there was no significant difference between the genders at 14 and, at 17, males outperformed females on all but visuo-spatial ability. This suggests gender-based developmental differences in terms of *g* and its sub-skills encapsulated in the Standard Matrices. With regard to verbal-analytic ability, at 17 males were found to have an advantage of 3.7 IQ points and this advantage increased into adulthood. Thus, if the Advanced Matrices

are a pure measure of *g* as some studies suggest, it would appear that adult “men have higher *g* than women” (Lynn *et al.*, 2004, p.412). Mackintosh and Bennett (2005) also found some evidence of a male advantage in their analysis of gender differences on the Advanced Matrices. They grouped the items according to the categories identified by Carpenter *et al.* (1990) and found differences only on certain items of the Advanced Matrices, with males performing significantly better than females on those items that require verbal-analytic reasoning in the form of an addition/subtraction rule (where a figure in one column is added to or subtracted from a figure in a second to produce the third) or a distribution of two rule (where two values of an attribute occur in each row, with the third value being null).

Despite the various disagreements over the constructs measured by the Advanced Matrices and the presence or absence of gender differences, it is consistently regarded as an excellent measure of individual differences in cognitive ability. The various Ravens Matrices have been widely used in South Africa under the assumption of being less biased than other, more verbal measures. However, this is based primarily on studies which utilized traditional methods of establishing bias, such as item difficulty and item discrimination (Owen, 1992; Rushton *et al.*, 2003). The limited research regarding the role of language and the varied evidence regarding the role of gender also highlight these two factors as important for further investigation. Consequently, the current study explored the relationships between the Advanced Matrices and measures of verbal-analytical ability and reading comprehension in young South African adults (average age 19.55 years), as well as how home language and gender affected these.

METHOD

The study followed a typical non-experimental design and a single set of cross-sectional measurements was obtained in relation to the variables of interest.

Participants

Participants were 100 undergraduate students. It was assumed that university students, having passed through the secondary education system, possessed a certain level of test-wiseness, as well as a minimum level of proficiency in English (the language of testing), ruling these out as extraneous variables. The gender and home language composition of the sample, shown in Table 1, reflects the population of first-year Psychology students at an English-medium, urban university in South Africa. Students were categorized as English first language (EL1) or English second language (EL2) based on their self report of the primary language spoken at home. Parental occupation was classified according to the Hall-Jones Scale of Occupational Prestige and used in conjunction with parental education to create a nominal classification of socio-economic status (Oppenheim, 1966).

Students participated on a voluntary basis, and written, informed consent was obtained following a verbal explanation of the purpose and requirements of the study.

Procedure

Participants completed the measures in a single session, following the standardized procedures for each measure. No time limits were imposed.

Instruments

The Raven's Advanced Progressive Matrices

The Advanced Matrices is the most difficult of the Ravens Progressive Matrices. Six of the twelve items in the training and familiarisation set (Set I), as well as the second set (Set II) of thirty-six items were utilized. Reliability and validity studies support the psychometric soundness of test scores for the Advanced Matrices (Paul, 1985; Raven *et al.*, 1998). Internal consistency studies have shown reliability estimates ranging from .60 to .98, with a median of .90 for the total score. Concurrent

validity evidence demonstrates correlates ranging from .50 to .70 between Advanced Matrices test scores and conventional tests of intelligence, such as the Wechsler and Stanford-Binet Scales (Strauss, Sherman, & Spreen, 2006).

Table 1. Breakdown of sample according to demographic factors

Demographic factor	Original responses (listed in order of magnitude, some single responses omitted)	Percentage
Age	Range: 16–69; Mode: 18 Mean: 19.55; SD: 5.50	
Gender	Male	29
	Female	71
Home Language	English 1st language	50
	English 2nd language: Zulu, Xhosa, Sepedi, Tswana, Sesotho, Swazi, Tsonga, Tshivenda	50
Race	White	34
	African, Indian, Coloured	66
Socio-economic status	Parental occupation: Levels 1–4 on the Hall-Jones Scale of Occupational Prestige; Parental education: above Matric	47
	Parental occupation: Levels 5–7 on the Hall-Jones Scale of Occupational Prestige; Parental education: below Matric	53
School leaving grades	Average symbol: A=1, B=2, C=3, D=4, E=5, F=6	1=12; 2=14; 3=35; 4=31; 5=6; 6=2
Psychology grades	75+ = 1 70–74 = 2 60–69 = 3 50–59 = 4 <50 = 5	1=17; 2=11; 3=25; 4=28; 5=19

Note: $N = 100$

The Similarities subtest of the South African Wechsler Adult Intelligence Scale (SAWAIS) This verbal subtest of the SAWAIS was designed to measure verbal concept-formation, associative thinking, ability to generalise and ability to draw abstractions (Spreen & Strauss, 1998). It is based on the Similarities subtest of the Wechsler Adult Intelligence Scale (WAIS) and consists of twelve pairs of words, where the participant must identify what the pairing has in common. The greater the level of abstraction in the given response, the higher the score assigned. The subtest is regarded as an effective measure of g , and was used here as a verbally-based measure of the same construct as the primarily non-verbal Advanced Matrices. Scores on the Similarities subtest appear unaffected by race and gender; although they can be influenced by level of education and age, and consistency estimates of 0.8 and above have been shown in previous research (Lezak *et al.*, 2004).

The Reading Comprehension subtest of the Stanford Diagnostic Reading Test (SDRT) — Blue Level

Employed within this study as a measure of English language comprehension, the Reading Comprehension subtest of the SDRT consists of a series of passages with multiple choice questions assessing the individual's understanding of the passage on two levels (literal and inferential). Literal comprehension questions assess a grasp of basic facts stated explicitly in the passage, while inferential comprehension questions evaluate the ability make deductions from given information. The passages increase in difficulty, draw on content from fiction, the social and natural sciences, history and human interest and are written at a level suitable for high school and first-year American college students

(Spreeen & Strauss, 1998).

The original version of the subtest contains nine passages and sixty items, which was unsuitable for the study due to time constraints. In order to preserve comprehension ability and the order of item difficulty while adjusting to an appropriate length, the third, sixth, seventh and ninth passages were omitted. Students thus answered thirty-two items based on five passages, fifteen of which assessed literal comprehension and seventeen of which assessed inferential comprehension. This adaptation, while suitable for the study, negated most of the previous psychometric information available on the subtest, particularly estimates of its scores' internal reliability, which had proved reasonable (.60 and higher) for the subtest as a whole (Spreeen & Strauss, 1998). It should be noted, however, that this type of adaptation is typically regarded as having minimal effect on the validity of the test as it involves only the removal of items designed to assess the same construct utilizing the same task format and therefore does not fundamentally change the scope of the task specification, merely the length (*cf.* Anastasi & Urbina, 1997; Murphy & Davidshofer, 2005)

Internal consistency reliability estimates for the three tests mentioned above for the sample as a whole were as follows: Advanced Matrices ($\alpha = .92$); Similarities ($\alpha = .66$) and SDRT ($\alpha = .79$). In addition to the above-mentioned measures, a brief questionnaire was devised to capture information regarding a number of pertinent demographic variables.

RESULTS

All analyses were based on raw Advanced Matrices scores, with each of the 36 items scored as either 0 (incorrect) or 1 (correct). The distributions of the scores were examined for their fit with the assumptions of parametric analyses by inspecting histograms, measures of central tendency and Kolmogorov-Smirnov tests of normality. Equality of variance checks were carried out only as needed in specific analyses, and the criterion of additive means was assumed in instances where the other assumptions had been met. The Advanced Matrices data were normally distributed, while both the Similarities and SDRT data were slightly skewed to the left. Consequently, both parametric and non-parametric analyses were initially run. Since the patterns of significance did not differ across the two sets of analyses, only the parametric results are reported.

Means and standard deviations for the raw scores on each measure are presented in Table 2. The variability of the SDRT scores was somewhat restricted as they manifested ceiling effects.

Table 2. Basic descriptive statistics for each measure

Variable	Mean	SD	Median	Mode	Range
Ravens Set II	18.26	8.51	18.5	27	1–34
Similarities	15.07	3.84	15	14	3–23
SDRT	26.67	4.62	28	30	8–32
Literal	12.86	2.07	13	13	5–15
Inferential	13.81	2.83	15	14	3–17

Note: $N = 100$; SDRT = Stanford Diagnostic Reading Test; Literal = Literal subcomponent of the SDRT; Inferential = Inferential subcomponent of the SDRT

A series of chi-squared tests was carried out to assess the relationship between the variables of gender and home language and other demographic variables known to influence test performance, namely ethnicity and socio-economic status (estimated by parental occupation and level of education). Estimates of significance on the basis of Yate's correction for continuity were also included, as the size of the final sample necessitated a conversion of many of the demographic variables to a dichotomous, nominal form, including home-language (English first language/English second language); ethnicity (white/black) and socioeconomic status (high/low).

There were no significant relationships between gender and any of the demographic variables, suggesting that it was possible to determine the influence of gender on test performance independently of these factors. Home language, on the other hand, was shown to be statistically significantly related to all of the demographic variables ($p < .0001$ in all cases), except gender ($p = .12$). This indicates that the variable of home language was closely interrelated with ethnicity ($\Phi = +.89$) and socio-economic status ($\Phi = -.70$).

In order to explore the relationships between general, literal and inferential English reading comprehension ability, performance on a verbal measure of g and performance on the Advanced Matrices, a series of Pearson Product Moment Correlation Coefficients were calculated. A moderate correlation was found between the Advanced Matrices and the Similarities subtest ($r = .57$; $p < .0001$), which is appropriate as a very high correlation would suggest excess duplication. The relationship between these measures appeared similar across genders ($r_{\text{males}} = .69$; $p < .0001$; $r_{\text{females}} = .65$; $p < .0001$) and Fisher's z test showed no significant difference between the correlations ($z = .52$; $p > .05$). The relationship between the Advanced Matrices and the Similarities subtest across the language groups appeared slightly more disparate ($r_{\text{EL1}} = .57$; $p < .0001$; $r_{\text{EL2}} = .51$; $p = .0002$), however the Fisher's z test revealed no significant difference between these sets of correlations ($z = .54$; $p > .05$). The Advanced Matrices thus appeared to assess verbal-analytical reasoning similarly between the gender and home language groups in relation to an independent measure of the construct, thereby suggesting a relatively high degree of construct comparability.

A moderate correlation was also found between the Advanced Matrices and the reading comprehension subtest ($r = .65$; $p < .0001$), with a very similar relationship between literal comprehension and the Advanced Matrices ($r = .62$; $p < .0001$) and inferential comprehension and the Advanced Matrices ($r = .61$; $p < .0001$). Fisher's z test indicated that the difference between these two correlations was non-significant ($z = -.02$; $p > .05$). The relationship between literal and inferential comprehension was reasonably strong ($r = .77$; $p < .0001$), but the lack of extreme strength in the relationship confirmed that they measured different aspects of reading comprehension.

Analyses of the relationship between the Advanced Matrices and English reading comprehension on the basis of gender and home language indicated that this relationship was very similar for males and females ($r_{\text{males}} = .65$; $p < .0001$; $r_{\text{females}} = .65$; $p < .0001$) and not significantly different ($z = -.01$; $p > .05$); while the relationship between performance on the two tests on the basis of home language was somewhat disparate ($r_{\text{EL1}} = .64$; $p < .0001$; $r_{\text{EL2}} = .52$; $p < .0001$), but also not significantly different ($z = 1.27$; $p > .05$). These results indicated that the relationship between performance on the Advanced Matrices and English reading comprehension ability did not differ on the basis of either gender or home language, thus suggesting there was no bias on these bases in this sample. Furthermore, the lack of distinction in the relationship between performance on the Advanced Matrices and literal or inferential comprehension suggested that both basic English ability and the ability to comprehend deeper, more abstract meaning in English affected Advanced Matrices performance to a similar extent.

The relationship between the reading comprehension subtest and the Similarities subtest was also moderate ($r = .57$; $p < .0001$), and statistically significant. This relationship was not anticipated, as given the verbal nature of both tests it was expected that it would be higher. However, it should be noted that this could have been a result of the high levels of performance (and thus ceiling effects) on the SDRT subtest in this sample. The relationship between literal comprehension and the Similarities subtest ($r = .49$; $p < .0001$) was somewhat lower than that between inferential comprehension and the Similarities subtest ($r = .57$; $p < .0001$), although the difference between the two was not statistically significant ($z = .77$; $p > .05$).

DISCUSSION

The results indicated a modest, but statistically significant relationship between verbal-analytic

ability, as assessed by the Similarities subtest, and the Advanced Matrices. This indicates that the tests shared some commonality in the constructs that they measured within the sample, but not excessively so, given that verbal-analytic ability is one of between two and three constructs tapped by the Advanced Matrices. More importantly, no statistically significant differences were found in the relationships between the two tests on the basis of either gender or home language, suggesting that neither of these variables substantially altered the observed associations between the two tests. This is despite the fact that several meta-analyses demonstrate a gender difference in cognitive abilities, as well as in the Advanced Matrices (Colom *et al.*, 2004; Lim, 1994; Lynn & Irwing, 2004). Given Lynn *et al.*'s (2004) finding of a male advantage on the verbal-analytic factor of the Standard Matrices that increased with age, we expected to find a similar difference in the Advanced Matrices' association with other measures of verbal-analytic ability. That this was not the case in the current study may have been related to the fact that the students were all from the Faculty of Humanities and were thus completing academic courses which demanded high levels of verbal ability, or more generally to the relatively homogenous nature of the sample. There has, however, also been recent interest in the possibility that gender differences in cognitive abilities are decreasing as a result of diminished effects of gender role stereotypes and other gender-differentiated environmental norms (Flynn, 1998).

Both literal and inferential English language comprehension as assessed by the SDRT were found to be related to performance on the Advanced Matrices, notwithstanding claims regarding the linguistic fairness of these (Owen, 1992; Paul, 1985; Raven *et al.*, 1998). The similarity of the relationship between literal comprehension scores and the Advanced Matrices ($r = .62$, $r^2 = 0.38$) and inferential comprehension and the Advanced Matrices ($r = .61$; $r^2 = 0.37$) was surprising, suggesting that although the two appear to measure distinct abilities, neither of these abilities was more likely than the other to determine Advanced Matrices performance. Furthermore neither of these predicted Advanced Matrices performance to any great extent at all, as each accounted for approximately 38% of the variance. This was unforeseen since the verbal-analytic reasoning factor identified on the Advanced Matrices (Carroll, 1993; McGrew & Flanagan, 1998) is defined as "the ability to start with stated rules, premises or conditions and engage in one or more steps to reach a solution to a problem" (McGrew & Flanagan, p. 15), while the inferential reasoning aspect of the reading comprehension subtest similarly assesses the individual's ability "to make inferences, draw conclusions, predict outcomes, evaluate situations, see cause and effect relationships, make comparisons and contrasts, understand characterization, verify the truthfulness or relevance statements and understand the author's purpose, bias, tone or mood ..." (Karlsen & Gardner, 1995, p.29). The similarity between the theoretical understandings of inferential ability, g and education as the ability to create meaning and infer relationships implies that those more successful at inferential comprehension should exhibit higher levels of g and vice-versa. However, the findings suggest that within this sample of South African university students, both literal and inferential comprehension ability played an equally important part in determining Advanced Matrices performance. There were, however, ceiling effects obtained on the reading comprehension subtest, suggesting that this test may have been too easy for some students, and may thus have been unable to distinguish nuances in English comprehension ability in the sample.

Although a more general concern regarding the relationship between English comprehension ability and the Advanced Matrices was thus raised, the relationship between the tests remained similar across genders and home languages, suggesting that neither of these variables acted as a source of bias on the Advanced Matrices in terms of the verbal construct validity. The relationship observed between the Similarities subtest and the reading comprehension subtest also raised some construct-related concerns, as it is reasonable to expect that a verbal comprehension test would rely quite heavily on verbal reasoning ability. The relationships between literal comprehension on the SDRT and the Similarities subtest and inferential comprehension and the Similarities subtest were similarly weak when separated in this way and showed no statistically significant differences. This provides

further evidence that the SDRT may not have been the most appropriate measure of English reading comprehension ability to use, particularly in adapted form, and it is strongly recommended that alternate measures be used to verify findings in future research.

The results obtained are also clearly limited by the small and specific nature of the sample, although utilizing participants with sufficient test-wiseness and English proficiency was necessary in practical terms. The homogenous nature of the sample should thus be kept in mind when considering the results; in addition, statistical power may not have been sufficiently high to provide sensitivity to small, but consistent differences. Since it was impossible to separate the effects of home language, ethnicity and socio-economic status in the sample, it is important to note that the home language variable was not a pure one and that the lack of statistically significant differences between the home language groups could reflect the confounded nature of this variable. In addition, the uneven division of male and female participants ($n_M = 29$ vs $n_F = 71$) indicates that results pertaining to gender should be interpreted with caution.

In conclusion, this study found that performance on the Advanced Matrices is statistically significantly related to the verbal-analytic constructs assessed in the Similarities subtest and to literal and inferential reading comprehension. Further, these relationships did not appear to differ across the variables of gender and home language, supporting the verbal-analytic construct validity of the test across these different groups. It is important to note that this does not suggest that either gender or home language could not act as potential sources of systematic differences in Advanced Matrices scores themselves.

While the absence of gender differences in the relationships merely adds further fuel to the conflicting results in this area, the findings regarding home language appear to be fairly novel, as minimal previous research has explored this aspect, with the exception of research by Raven *et al.* (1998), which suggested systematic differences in performance on the Advanced Matrices between Dutch and French speakers. The conflicting results obtained here provide further evidence of the necessity of additional construct validation of the Advanced Matrices, particularly in multi-lingual contexts such as South Africa. In addition, this study represents research on the Advanced Matrices in the South African context that attempts to move beyond ethnicity as an all-encompassing source of bias, and to examine other, equally salient possibilities.

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